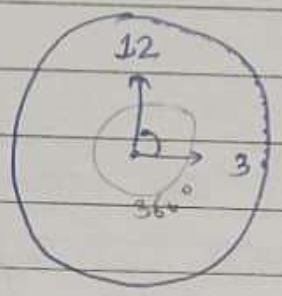
1344. Angle b/w hands of a clock

0. Analog clock

$1 <= hour <= 12$

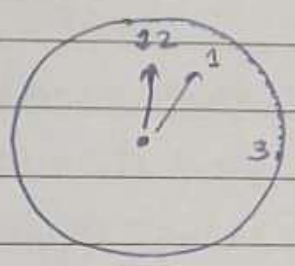
$0 <= min \leq 59$



hr min
3:00

hour

Hour



A full circle = 360°

clock has 12 hours

$$\boxed{1 \text{ hour} = \frac{360}{12} = 30^\circ / \text{hour}}$$

~~Minute hand:~~

~~moves 360° in 60 minutes~~

Hour hand:

moves 30° per hour

also moves while minutes pass

$$\boxed{\frac{30^\circ}{60} = \frac{1}{2}^\circ \text{ per min} \Rightarrow 0.5^\circ \text{ per minute}}$$

Therefore,

hour hand → 30° / hour

hour hand → 0.5° / min

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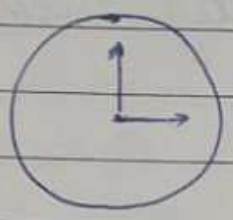
Minute hand

→ 0°/hour
→ 6°/min

moves 360° or 0° per hour



- Moves 360° in 60 minutes

$$\frac{360}{60} = 6° \text{ per min}$$

eg: x hour y minute

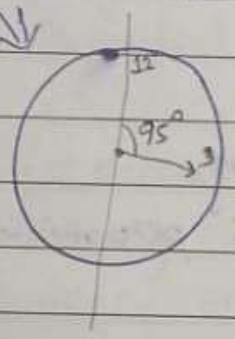
Hour $= 30 \times x + \frac{1}{2} \times y \Rightarrow \boxed{\frac{60 \times x + 1 \times y}{2}}$

minute $= \boxed{x \times 0 + y \times 6}$

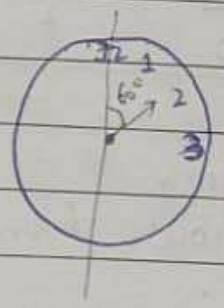
eg: 3 hour 10 min

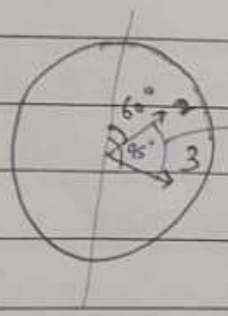
hour $= \frac{60 \times 3 + 10 \times 1}{2} = \frac{180 + 10}{2} = \frac{190}{2} = 95°$

⇓



min $= 3 \times 0 + 6 \times 10 = 60 \Rightarrow$

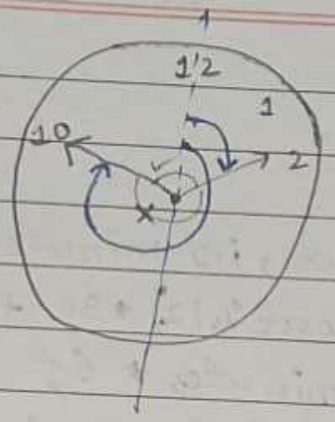




→ difference ⇒ $95° - 60° = 35°$

|hour angle − minute angle|

eg



1:50

if angle > 180°
then 360° − angle

small angle + big angle = 360°
Small angle = 360° − Big angle

why 360° − angle?
because clocks form two angles (one small, one large)
we always return the smaller one (≤ 180°)

eg Time: 1:50

hour angle = 1 × 30 + 50 × 0.5 = 55°
minute angle = 50 × 6 = 300°
difference = |55 − 300| = 245°
Smallest angle = 360 − 245 = 115°
Ans = 115°

we write ⇒
hour angle = (hour % 12 * 60 + minutes) * 0.5
clock resets after 12 hours, not 24.
If we write only ⇒ (hour * 60 + minutes) * 0.5 ⇒ angle becomes wrong

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Code

```
Class Solution {
public:
    double angleClock (int hour, int minutes) {
        double hourAngle = (hour % 12) * 30 + minutes * 0.5;
        double minuteAngle = minutes * 6;
        double angle = abs (hourAngle - minuteAngle);
        if (angle <= 180)
            return angle;
        else
            return 360° - angle;
    }
};
```